**Q. 1** (Yogurt). a. Let $R$ be the event that you get a rash on a given day. Also let $E_1$, $E_2$, and $E_3$ be the events that you are served blueberry, banana and jalapeno yogurt respectively on a given day. We are asked to calculate $\mathbf{P}(R)$. We can use the law of total probability to calculate this: $\mathbf{P}(R) = \sum_{i=1}^{3} \mathbf{P}(R|E_i)\mathbf{P}(E_i)$.
According to the given information we have:
$\mathbf{P}(E_1) = 0.2, \mathbf{P}(E_2) = 0.5, \mathbf{P}(E_3) = 0.3,$
$\mathbf{P}(R|E_1) = 0, \mathbf{P}(R|E_2) = 0.3, \mathbf{P}(R|E_3) = 0.$
When we substitute the values, we get $\mathbf{P}(R) = 0.5 \cdot 0.3 = 0.15$.

b. Let $NR$ be the event that you don't get a rash on a given day. We are asked to calculate $\mathbf{P}(E_2|NR)$. We can use the Bayes' Theorem as follows:
$\mathbf{P}(E_2|NR) = \frac{\mathbf{P}(NR|E_2)\mathbf{P}(E_2)}{\mathbf{P}(NR)}$.
We know $\mathbf{P}(E_2) = 0.5$ and $\mathbf{P}(NR|E_2) = 1 - \mathbf{P}(R|E_2) = 0.7$.
Since we already calculated $\mathbf{P}(R)$ in part a, we can directly say:
$\mathbf{P}(NR) = 1 - \mathbf{P}(R) = 1 - 0.15 = 0.85$ since $NR$ and $R$ are complementary events.
So, we have:
$\mathbf{P}(E_2|NR) = \frac{0.7 \cdot 0.5}{0.85} = 0.412$.

c. Let $N$ be the number of times during one week that you eat banana yogurt, but do not get a rash. We are asked to find $\mathbf{E}[N]$.
Define $N_i$ for $i$ in $\{1, 2, \ldots, 7\}$ as:

$$N_i = \begin{cases} 1 & \text{if you eat banana yogurt and do not get rash on day } i \\ 0 & \text{otherwise} \end{cases}$$

Note that we have: $N = \sum_{i=1}^{7} N_i$.
And by the linearity of expectation, we have: $\mathbf{E}[N] = \sum_{i=1}^{7} \mathbf{E}[N_i]$.
The $N_i$'s are Bernoulli random variables with the same probability $p$ of success. The parameter, $p$, of this random variable is equal to the probability of eating a banana yogurt and not getting a rash on any given day.
So $p = \mathbf{P}(NR|E_2)\mathbf{P}(E_2) = 0.7 \cdot 0.5 = 0.35$.
$\mathbf{E}[N_i]$ is therefore equal to 0.35 as well.
Finally, $\mathbf{E}[N] = \sum_{i=1}^{7} 0.35 = 2.45$.

**Grading comments for Q. 1**

- In general, the performance on this question is pretty good. Almost everyone got parts a. and b. completely correct.
- Part c:
  - I think a handful of people didn't really understand what was asked in part c. Some people used the result of the second part as the single day probability in part c., which is incorrect. Also, I think some people solved the

question as if we are eating banana yogurt everyday, which is not a given in this question. You have to include the probability of eating a banana yogurt in your expectation calculations as well.

  - Some people defined a binomial distribution for part c. and used the expected value formula of the binomial distribution to solve the question. Although this gives the correct answer, the reasoning is incorrect since the days are not given as independent in the question, which would be required to define a binomial distribution. Therefore, some students lost points due to this incorrect reasoning.
  - There are also some students who have done the linearity of expectation calculations, but using the independence of the days as their reasoning. Since this is a wrong claim, they have lost some points as well. The linearity of expectation does not require independence.

**Q. 2** (Snowfall). *Note: You can understand 'more than 1 foot' and 'between 1 and 2 feet' as including or not including the boundaries. In fact, this will not affect the answers, since the probability of the random variables lying on the boundaries will be 0.*

a. $\mathbf{P}(S_P \geq 1) = \mathbf{P}(S_P \geq 1, S_N \geq 0) = \frac{1}{2}$.

b. We have $\mathbf{P}(S_N \geq 1) = \mathbf{P}(S_N \geq 1, S_P \geq 0) = \frac{1}{2}$ and $\mathbf{P}(S_N \geq 1, S_P \geq 1) = \frac{1}{3}$. Therefore,
$$\mathbf{P}(S_N \geq 1, S_P \geq 1) \neq \mathbf{P}(S_P \geq 1)\mathbf{P}(S_N \geq 1).$$
This implies that the two events are not independent.

c. For any $x, y \geq 0$, define the event $A_{xy} \triangleq \{S_P \geq x, S_N \geq y\}$. Moreover, define $A^* = \{S_P \in [1,2), S_N \in [1,2)\}$. Then $A^* = A_{11} \cap A^c$, where $A = A_{12} \cup A_{21} \subset A_{11}$. Thus, we have
$$\begin{aligned}
\mathbf{P}(A^*) &= \mathbf{P}(A_{11}) - \mathbf{P}(A) \\
&= \mathbf{P}(A_{11}) - \mathbf{P}(A_{12} \cup A_{21}) \\
&= \mathbf{P}(A_{11}) - [\mathbf{P}(A_{12}) + \mathbf{P}(A_{21}) - \mathbf{P}(A_{12} \cap A_{21})] \\
&= \mathbf{P}(A_{11}) - [\mathbf{P}(A_{12}) + \mathbf{P}(A_{21}) - \mathbf{P}(A_{22})] \\
&= 1/3 - [1/4 + 1/4 - 1/5] \\
&= 1/30 = 0.033.
\end{aligned}$$

**Grading comments for Q. 2**

- Part a: The most common mistake for this part is failure to identify the relationship between $\mathbf{P}(S_P \geq 1)$ and $\mathbf{P}(S_P \geq s, S_N \geq t)$. In fact, many students start their solution by expressions such as $\mathbf{P}(S_P \geq 1) = \sum_{t \geq 0} \mathbf{P}(S_P \geq 1, S_N \geq t)$, which lead to erroneous results. I think it's helpful if we can better understand the relationship between joint distribution and marginal distribution.

- Part b: There should be little problem to solve this part given that $\mathbf{P}(S_N \geq 1)$ and $\mathbf{P}(S_P \geq 1)$ are computed correctly. Since we're answering a probability question, please be sure to verify independence of events from the definition. Unfortunately, it's not sufficient to simply argue that 'Princeton and New York are geographically close to each other'.

- Part c: The most common mistake for this part is to think

  $$\mathbf{P}(S_P \in [1,2), S_N \in [1,2)) = \mathbf{P}(S_P \geq 1, S_N \geq 1) - \mathbf{P}(S_P \geq 2, S_N \geq 2).$$

  In fact, the event $\{S_P \geq 1, S_N \geq 1\}$ is larger than the union of $\{S_P \geq 2, S_N \geq 2\}$ and $\{S_P \in [1,2), S_N \in [1,2)\}$. Intuitively, if $S_P \geq 1, S_N \geq 1$, it might also be the case that $S_P \geq 2, S_N \in [1,2)$, or $S_N \geq 2, S_P \in [1,2)$.

  There are also a few students writing

  $$\mathbf{P}(S_P \in [1,2), S_N \in [1,2)) = \mathbf{P}(S_P \in [1,2))\mathbf{P}(S_N \in [1,2)),$$

  which is not the case either, since $S_P$ and $S_N$ are not independent.

**Q. 3** (COVID vaccine)**.** a. In seven days, the number of vaccine you get follows the Binomial distribution with parameter $(7, 0.05)$ by independence of the days, so the probability of not getting the vaccine in the first week is $p_1 = \binom{7}{0}(1-0.05)^7$. Similarly, the times that you get exposed to COVID in seven days follows the Binomial distribution with parameter $(7, 0.01)$, so the probability that you get exposed to COVID three times is $p_2 = \binom{7}{3}0.01^3(1-0.01)^4$. Finally, since COVID exposition and availability of vaccination slots are independent, the final probability is $p_3 = p_1 \cdot p_2$.

b. We let $T_1, T_2$ be the arrival time for the first and second dose of vaccine. From the lecture notes, we know both $T_1, T_2 - T_1$ follow Geometric distribution with parameter $0.05$. So we have $\mathbf{E}[T_2] = \mathbf{E}[T_1] + \mathbf{E}[T_2 - T_1] = \frac{2}{0.05} = 40$. *Note: From the lecture notes, we also know that $T_1$ and $T_2 - T_1$ are independent.*

c. We let $X_i$ be indicator random variable of whether one gets exposed to COVID on the $i$-th day, i.e. $\mathbf{P}(X_i = 1) = 0.01$ and $\mathbf{P}(X_i = 0) = 0.99$. In addition, we let $N$ be the first date that one receives the first dose of the vaccine, we know $N \sim Geo(0.05)$. We let $A$ be the event that one doesn't get exposed to COVID before one has the first dose of vaccine. And our goal is to compute $1 - \mathbf{P}(A)$. We have

$$\begin{aligned}
\mathbf{P}(A) &= \sum_{n=1}^{\infty} \mathbf{P}(\sum_{i=1}^{n-1} X_i = 0 | N = n)\mathbf{P}(N = n) \\
&= \sum_{n=1}^{\infty} \binom{n-1}{0} 0.99^{n-1} \cdot 0.05 \cdot 0.95^{n-1} \\
&= \frac{0.05}{1 - 0.95 \cdot 0.99}
\end{aligned}$$

The term $\sum_{i=1}^{n-1} X_i = 0$ means one doesn't get expose to COVID before the $n$-th day, which is the date that one gets the first vaccine. The second equation follows from the independence between $X_i$ and $N$, and the third equation is achieved by the geometric series formula. Thus, the final probability is $1 - \mathbf{P}(A) = 1 - \frac{0.05}{1-0.95 \cdot 0.99}$.

**Grading comments for Q. 3**

- Parts a and b: Most students did a. and b. correctly. And the existing issue mostly involves not figuring out correct distributions (e.g. Binomial or Geometric distribution) of defined random variables.
- Part c:
  - Correctly formulating this problem into math language seems challenging, so briefly describing your intuition (although with wrong method) gets some partial credits.
  - Some students fixed the date that one receives the first vaccine, and get the probability with that fixed $N = n$. However, here $N$ is also a random variable, treating it as a fixed number is not reasonable. So in this case, $4.0$ points were subtracted for not using the Law of total probability.
  - There are also some cases that students correctly formulate the problem, but have some computation issues. In this case, $2.0$ or $1.0$ points are subtracted depending where you get your computation issue.